

LABORATORY OF AUTOMATION SYSTEMS

Davide Samorì

Dipartimento di Ingegneria dell'Energia Elettrica e dell'Informazione (DEI)

Università di Bologna

Email: davide.samori@unibo.it

A QUICK OVERVIEW OF OUR LESSONS

PC LAB5

Username: **Studente**
Password: **DEI01**

DC MOTOR MODEL

Using Laplace transformation, it results:

[1] $V_A(s) = R_A I_A(s) + s L_A I_A(s) + V_M(s)$

[2] $C_m(s) = k_m I_A(s)$

[3] $s J \Omega_m(s) = C_m(s) - b \Omega_m(s)$

[4] $V_M(s) = k_m \Omega_m(s)$

The two dynamics are then expressed by the transfer functions:

[5] $I_A(s) = \frac{V_A(s) - V_M(s)}{R_A + s L_A}$

[6] $\frac{\Omega_m(s)}{C(s)} = \frac{1}{b + s J}$

Slide Prof. Melchiorri
LAS_03_Motion.pdf
Pag. 10

$$\cancel{C(s)} \frac{\Omega_m(s)}{\cancel{C(s)}} = \frac{1}{b + sJ} C(s) \quad \Rightarrow$$

$$\Omega_m(s) = \frac{C(s)}{b + sJ}$$

$$I_A(s) = \frac{V_A(s) - V_M(s)}{R_A + sL_A}$$

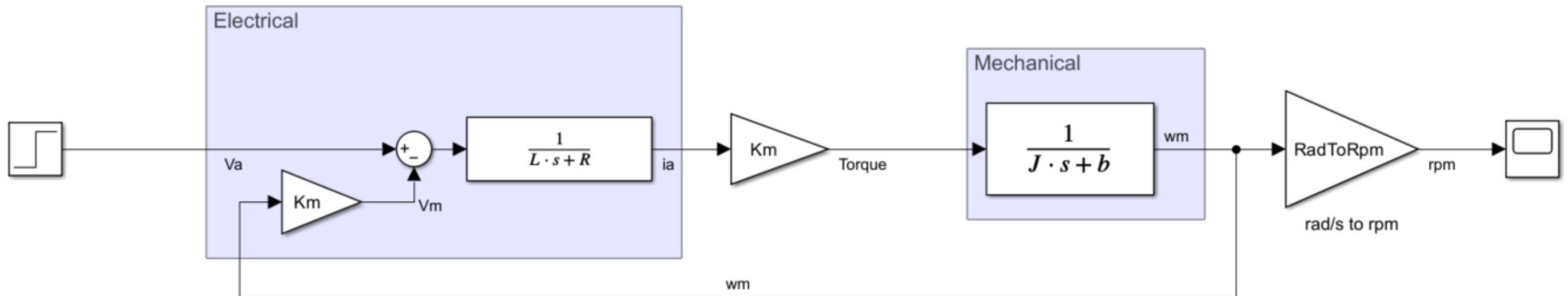
$$= I_a(s) \cdot k_m \cdot \frac{1}{b + sJ}$$

$$C_m(s) = k_m I_A(s)$$

$$= \frac{V_A(s) - V_M(s)}{R_A + sL_A} \cdot k_m \cdot \frac{1}{b + sJ}$$

$$V_M(s) = k_m \Omega_m(s)$$

$$\Omega_m(s) = \left(V_A(s) - k_m \cdot \Omega_m(s) \right) \cdot \frac{1}{R_A + sL_A} \cdot k_m \cdot \frac{1}{b + sJ}$$



DATI MOTORE Motor ratings	SIMBOLI Symbols	UNITA' Units	PMB 1524
COPPIA ALLA VELOCITA' NOMINALE Torque at rated speed	Cn	Nm	0.22
VELOCITA' NOMINALE Rated speed	Nn	RPM	1500
POTENZA NOMINALE Rated output	Pu	W	35
TENSIONE NOMINALE Rated voltage	Vn	V	24
CORRENTE NOMINALE Rated current	In	A	1.9
VELOCITA' A VUOTO Idle speed	Nv	RPM	1750
TENSIONE A VUOTO Idle voltage	Vv	V	24
CORRENTE A VUOTO Idle current	Iv	A	0.3
F.C.E.M ALLA VELOCITA' NOMINALE B.E.M.F at rated speed	E	V	19.5
COPPIA DI PICCO Peak torque	Cp	Nm	0.88
CORRENTE DI PICCO Peak current	Ip	A	8
MAX VELOCITA' Max speed	N max	RPM	4000
RENDIMENTO Efficiency	η	%	73
DATI MECCANICI Mechanical data			
MOMENTO D'INERZIA ROTORICO Rotor inertia	J	Kgcm ²	0.352
MAX. ACCELERAZ. TEORICA Max theoretical acceleration	α	rad/sec ²	25000
MAX TENSIONE Max voltage	V max	V	65
COSTANTE DI TEMPO MECCANICA Mechanical time constant	Tm	ms	85
VENTILAZIONE Ventilation			Naturale
GRADO DI PROTEZIONE Protection (IEC 34.5)		IP	54
PESO Weight	G	Kg	1.35
DATI ELETTRICI Electric data			
COSTANTE DI TENSIONE Voltage constant	Ke	Vs/rad	0.12
COSTANTE DI COPPIA Torque constant	Kt	Nm/A	0.11
COSTANTE DI TEMPO TERMICA Thermal time constant	Tt	min	15
COSTANTE DI TEMPO ELETT. Electrical time constant	Te	ms	1.3
RESISTENZA D'ARMATURA Armature resistance	Ra	Ohm	--
RESISTENZA D'ARMATURA CON SPAZZOLE Terminal resistance	Rm	Ohm	3.20
INDUTTANZA D'ARMATURA Armature inductance	La	mH	4.1

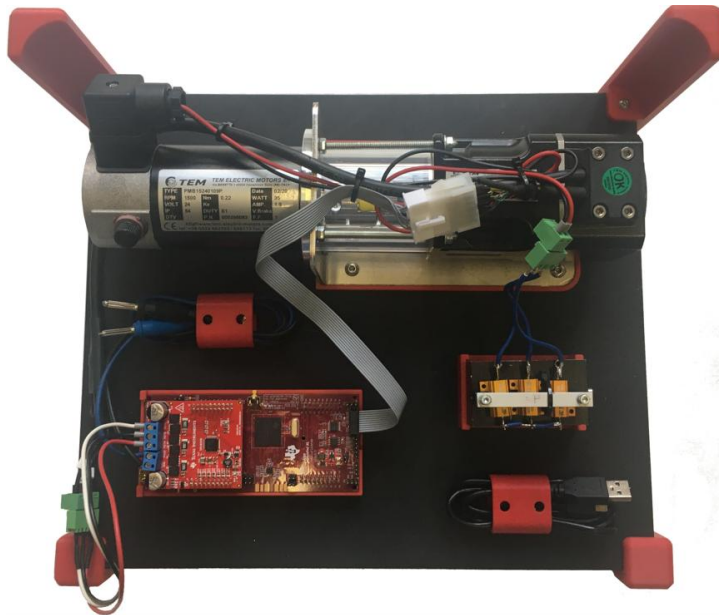
understand and find the motor parameters in the model

WORK WITH ACQUIRED DATA

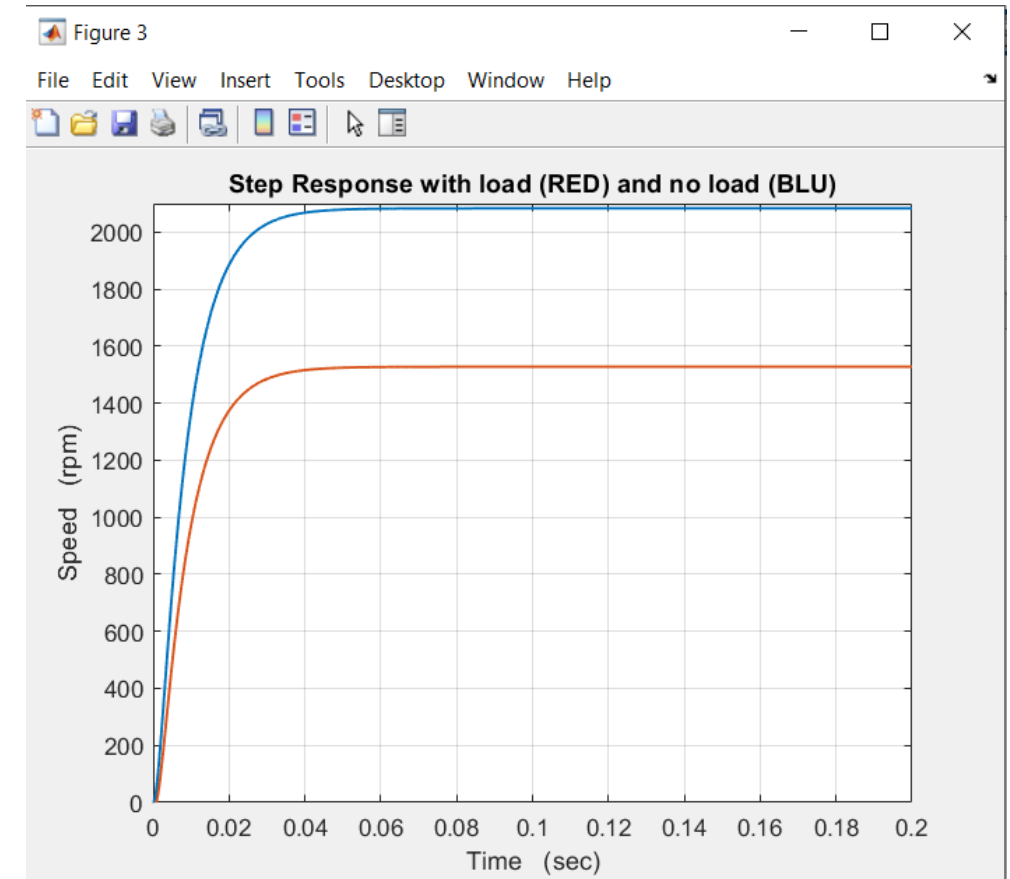
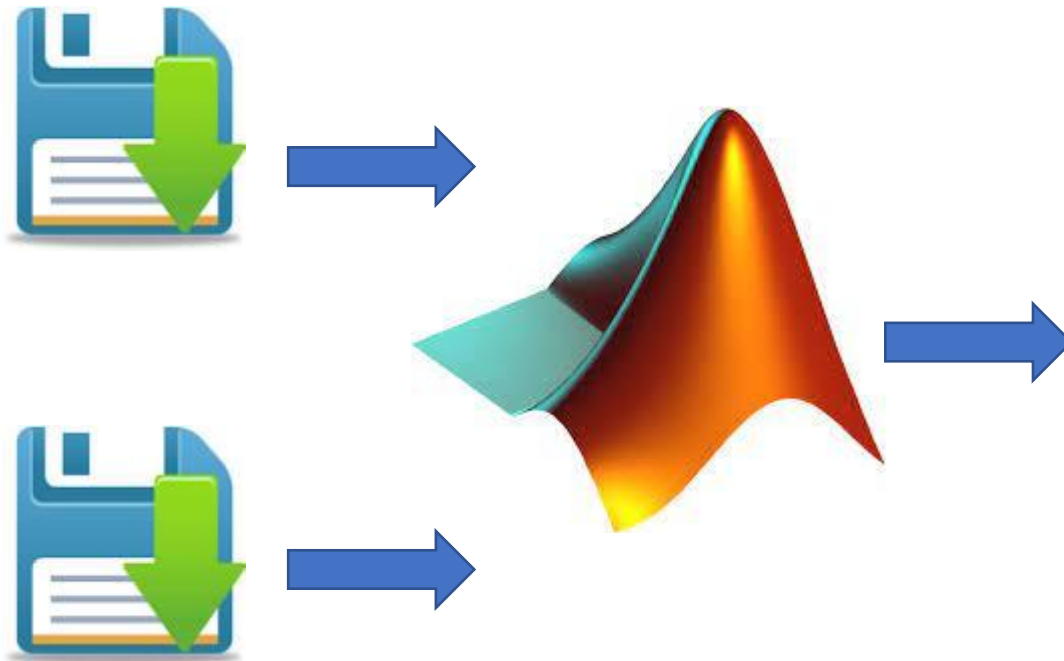
It is often useful to work with previously acquired data:

- compare different control strategy
 - In the setup we can only run one control algorithm at a time
 - In Simulink you can run two or more control algorithm
- Post data processing (Eg. Time constant, overshoot, ecc...)
- Compare real data with simulation data
- Ecc...

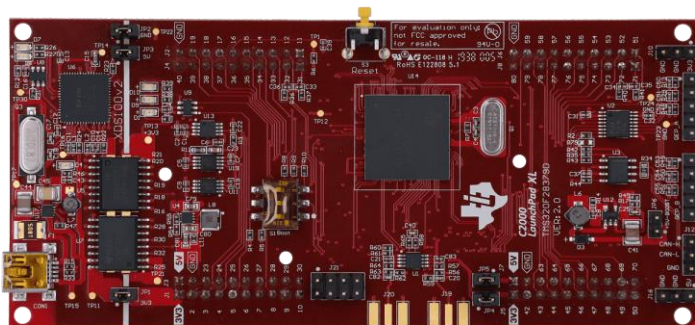
HOW TO STORE DATA



HOW TO PLOT DATA

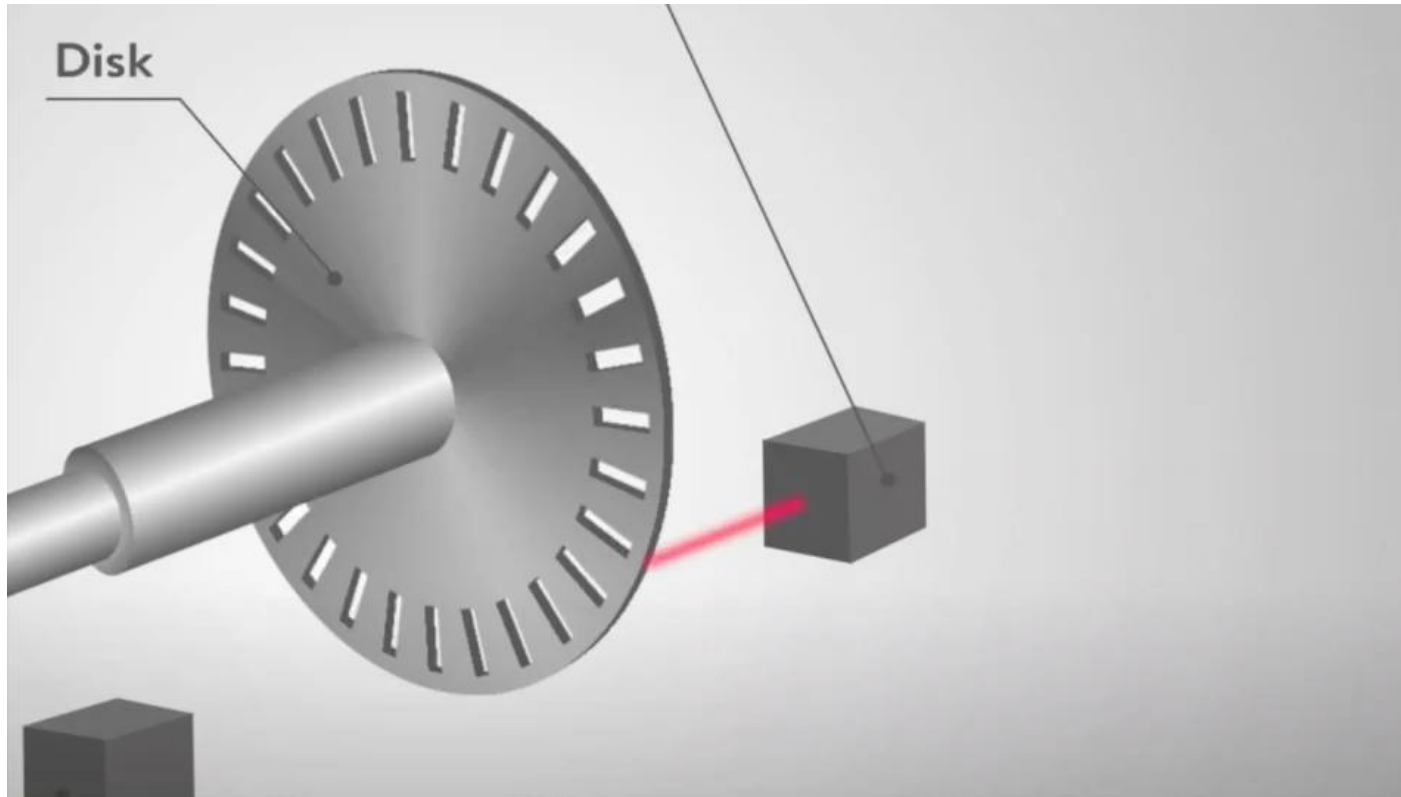


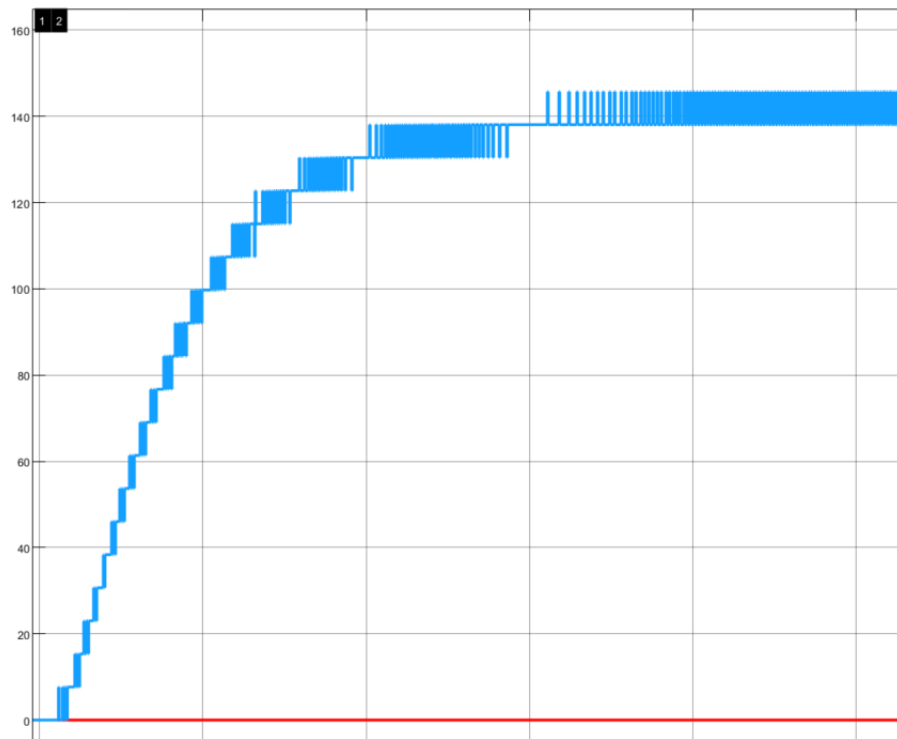
In order to drive our motor, What physical quantities we need to measure?



SPEED/POSITION

When the motor rotates, we have an alternation of light/dark. Higher speed, higher frequency

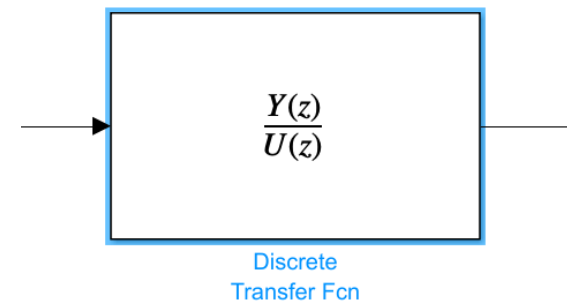


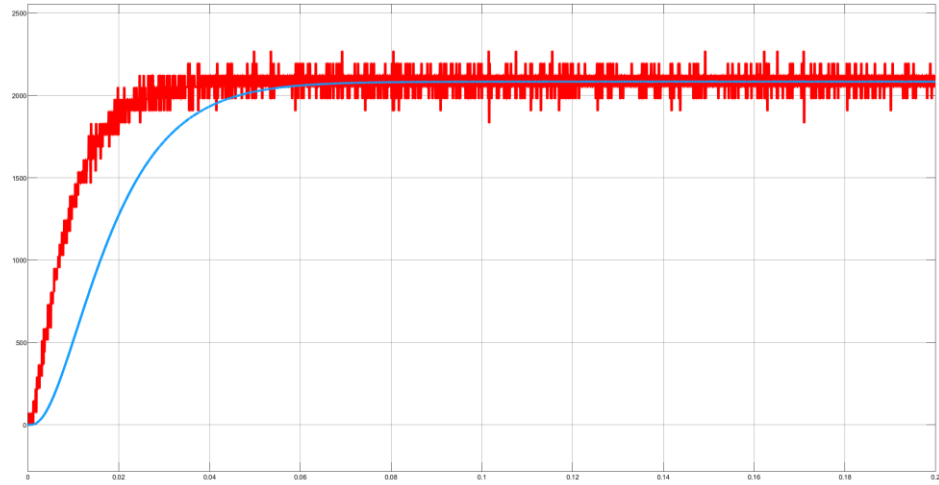
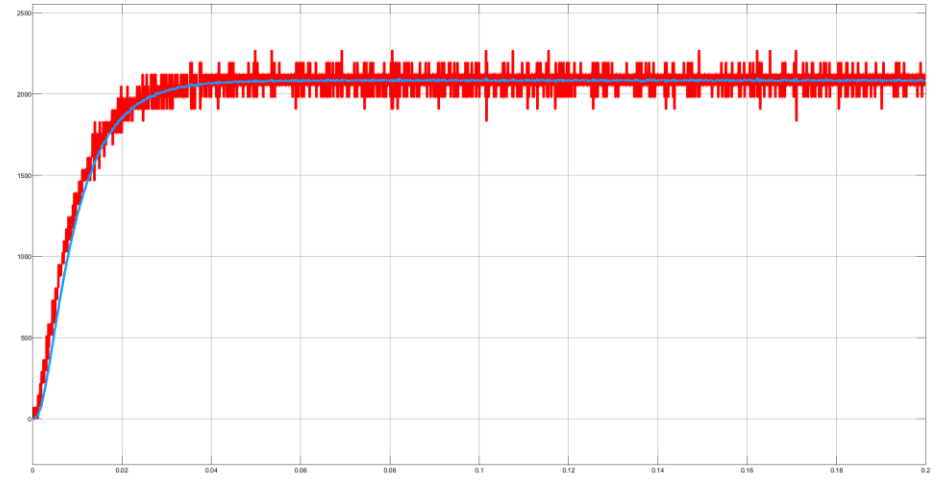
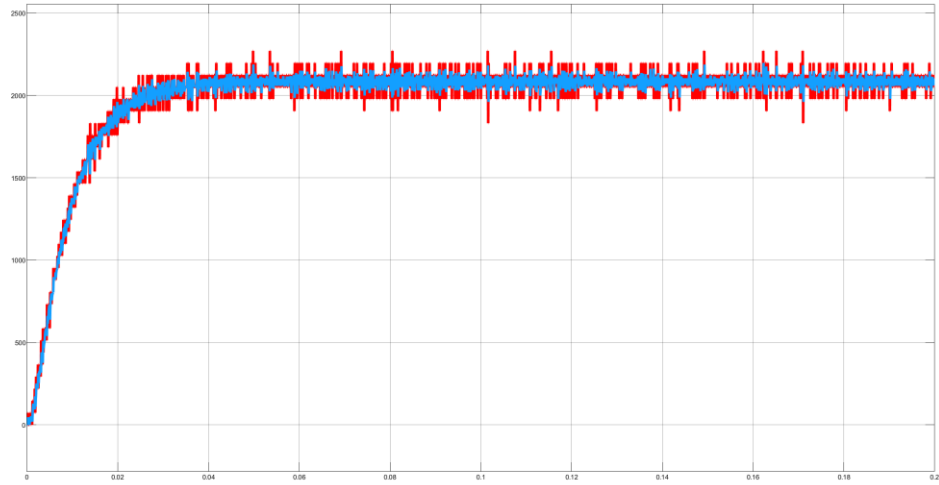


We need a filter:

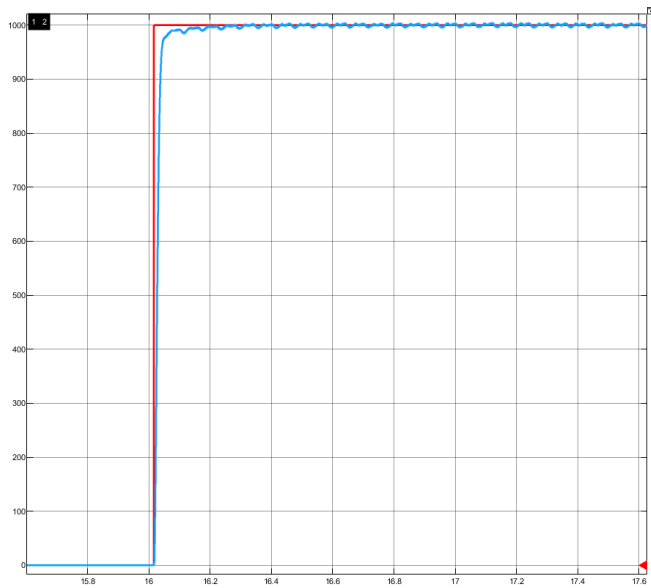
DO YOU KNOW ANY FILTER?

$$G(s) = \frac{1}{1 + s\tau}$$

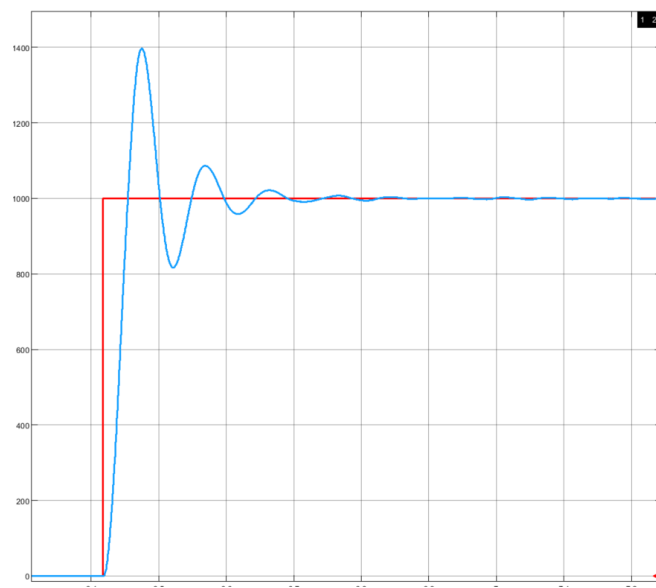




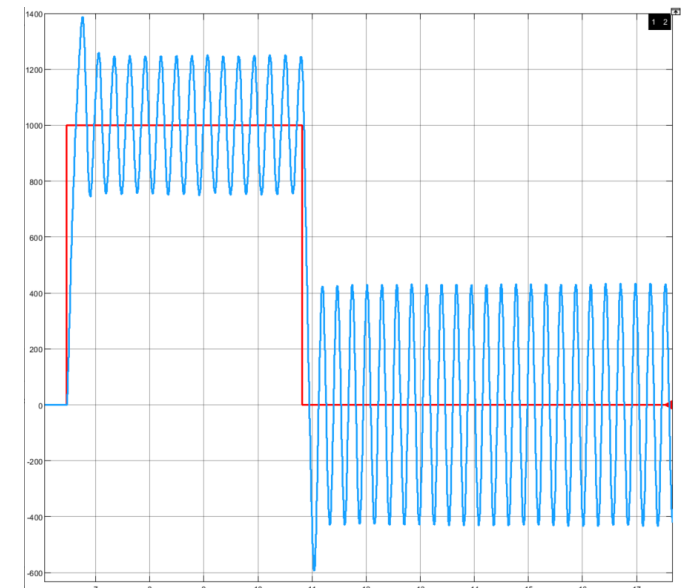
- test different value of TaoOmegaFilt.



% Omega filter
TaoFilterOmega=0.002;



% Omega filter
TaoFilterOmega=0.02;



% Omega filter
TaoFilterOmega=0.2;

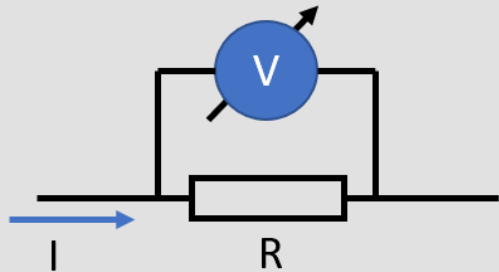
CURRENT SENSING

If we want to measure a current with our microcontroller, we must convert it into a voltage.

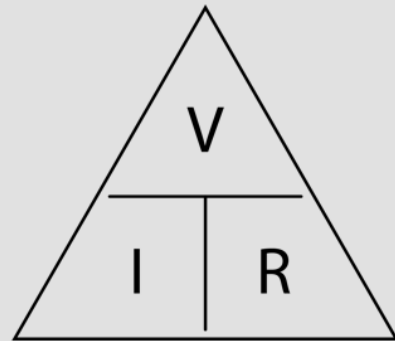
We can divide the measurement methods into two groups:

SHUNT RESISTOR

A current flow on a resistor and we measure the voltage dropout.



$$I = \frac{V}{R}$$



OHM Law

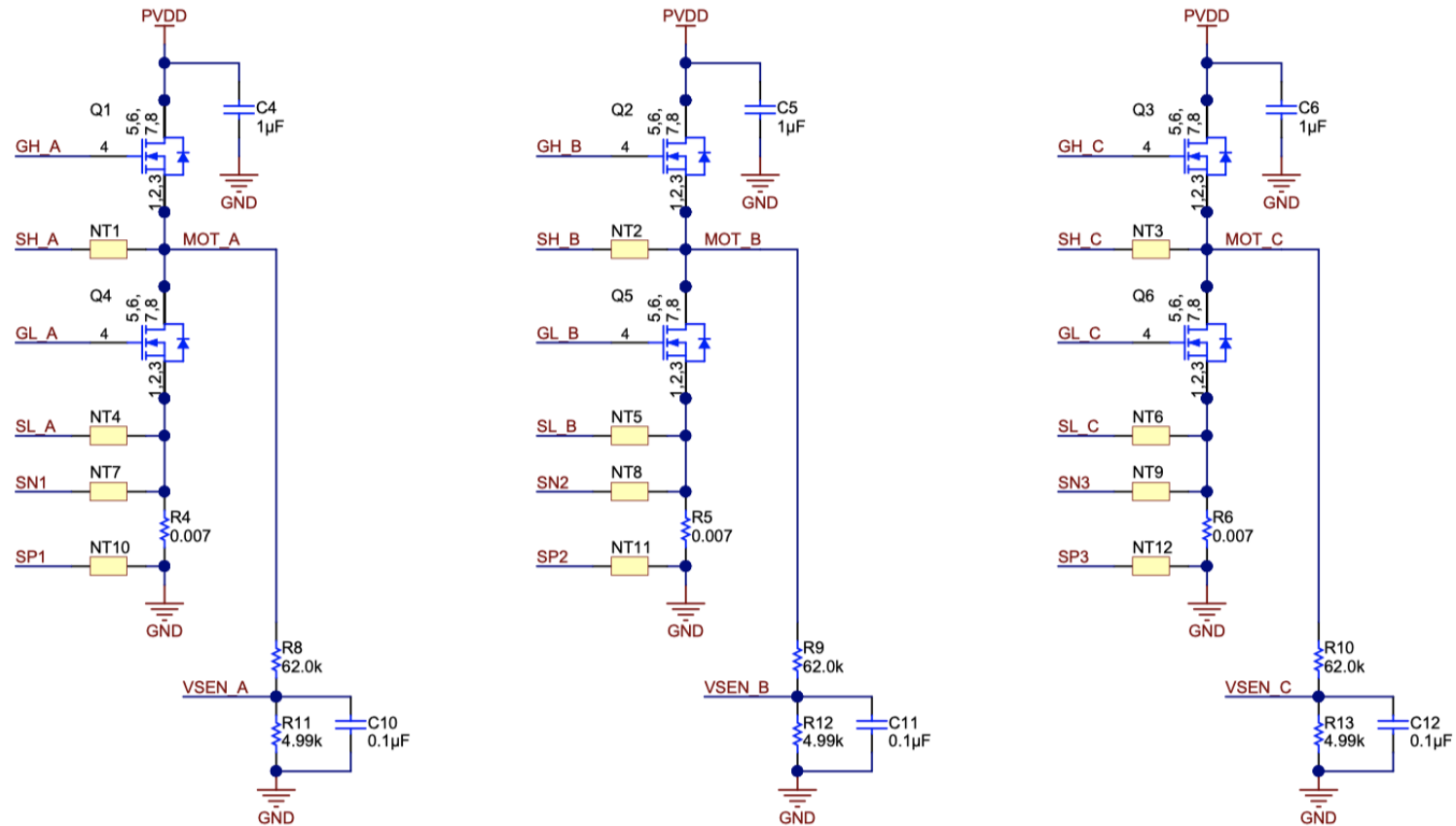
Current

“ELECTROMAGNETIC FIELD” SENSOR



€€€€€

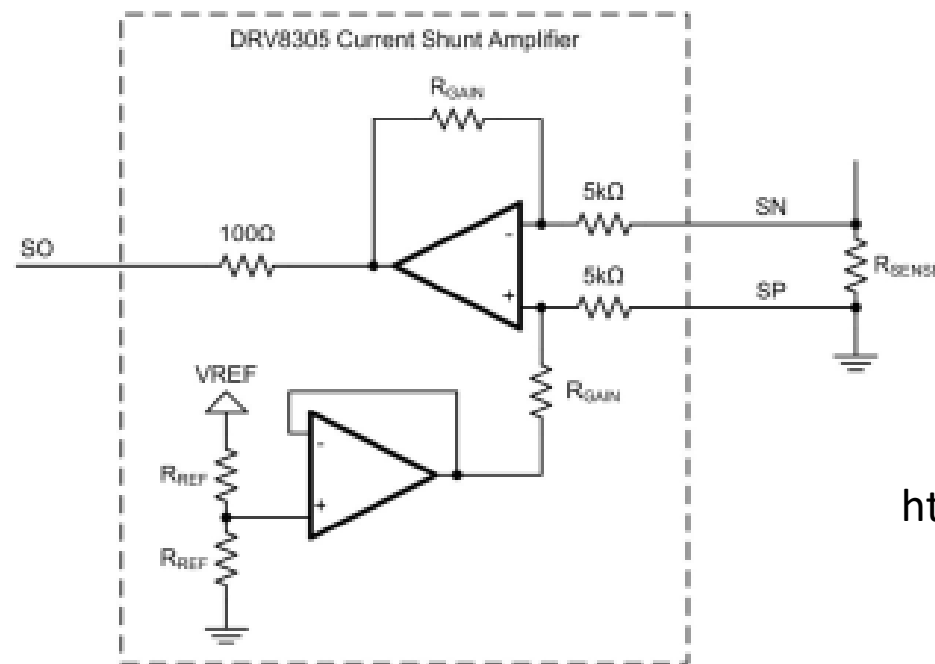
HALF-BRIDGES & BACK-EMF SENSE



CURRENT SENSING – OUR BOARD

5.2 Low-Side Current Shunt Sense

The BoosterPack has low-side current shunt sense for each half-bridge (phases A, B, and C). The current sense setup takes advantage of the DRV8305's triple shunt current amplifiers (phases A, B, and C). The configuration for the low-side sense is shown in Figure 9. The differential amplifier senses voltage across a 0.007- Ω power sense resistor with differential connections. The differential voltage is then amplified by 10 V/V and centered at 1.65 V to allow for sensing both positive and negative currents. The sense resistor has been scaled for 0- to 20-A peak currents.

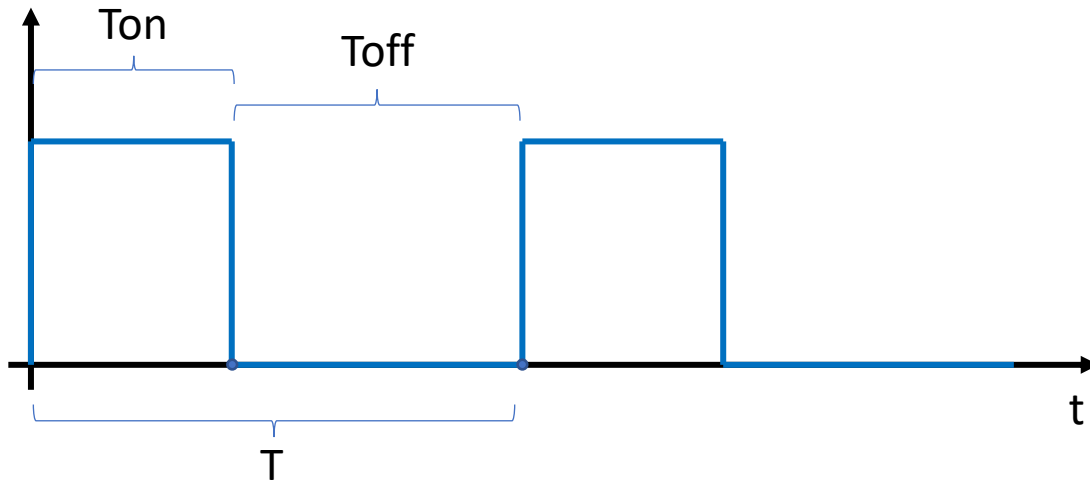


<https://www.ti.com/lit/pdf/slvuai8>

Pulse Width Modulation (PWM)

Pulse Width Modulation, commonly known as PWM, is a way to regulate the average power delivered to the motor by switching a power signal ON and OFF similar to a square wave.

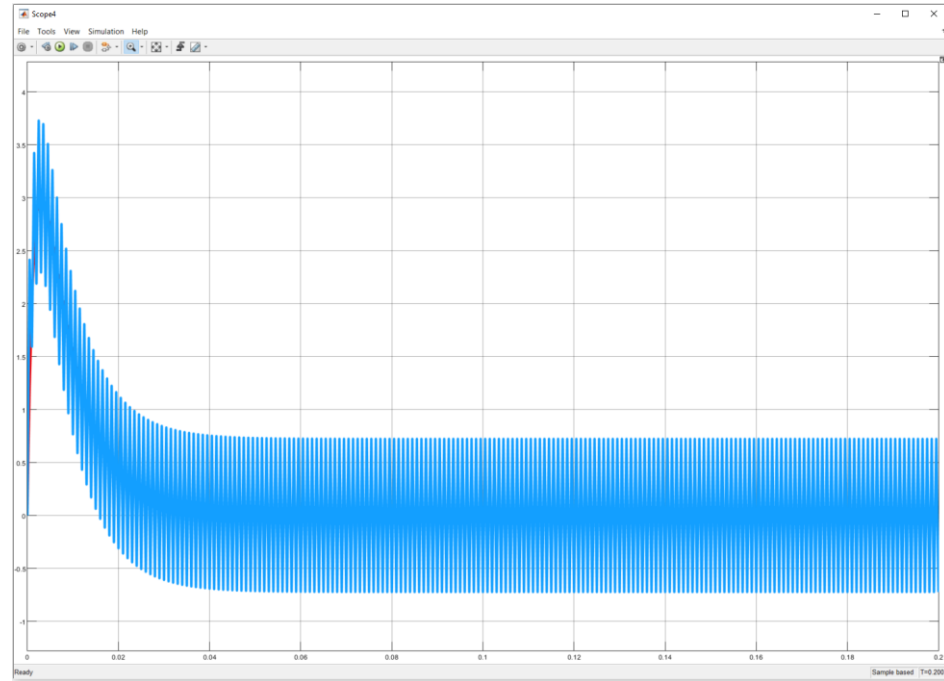
In a PWM signal we have two parameters: **Frequency** and **duty-cycle**



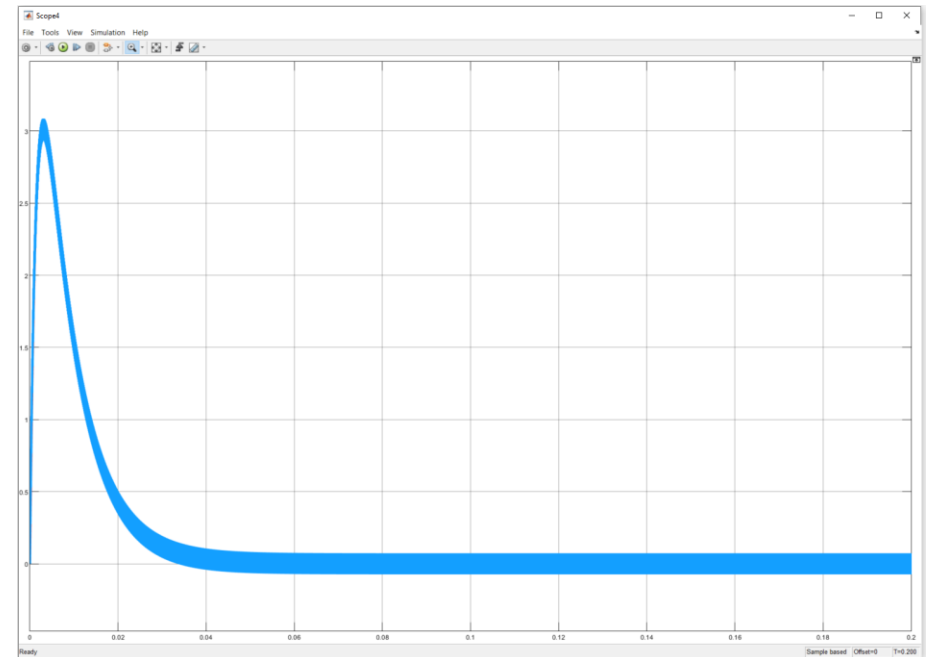
$$period = T_{on} + T_{off} = T$$

$$frequency = \frac{1}{T}$$

$$duty cycle = \frac{T_{on}}{T_{on} + T_{off}} * 100 = \frac{T_{on}}{T} * 100$$



PWM frequency



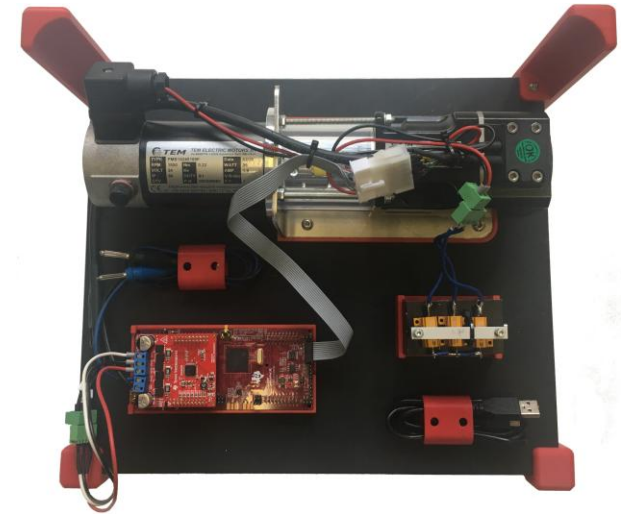
HOW TO CONNECT



1: Power Supply



2: Power Supply cable



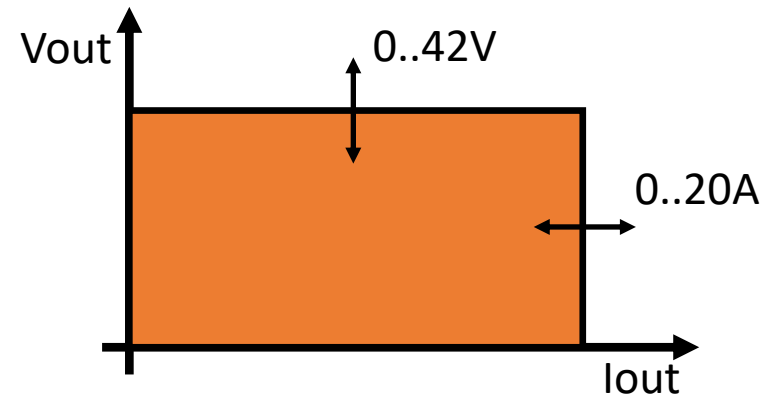
3: Setup

POWER SUPPLY



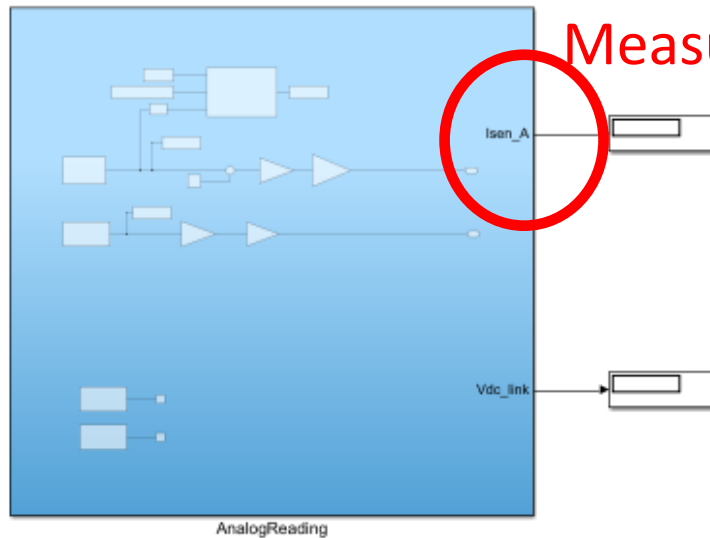
It can be used as:

- ideal voltage generators
- ideal current generators



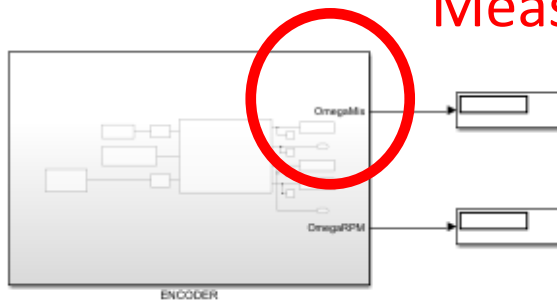
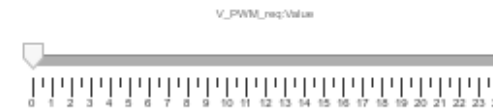
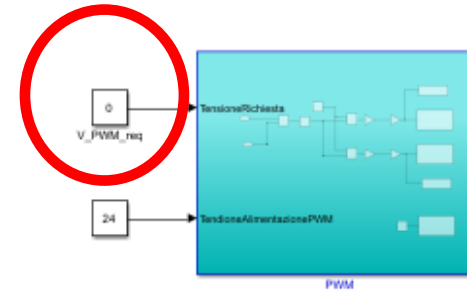
$P_{max}=320W$

When used as voltage generators, setting an appropriate current, it is possible to "protect" the load from overcurrent or short circuits.



Measured Current

Voltage to the motor



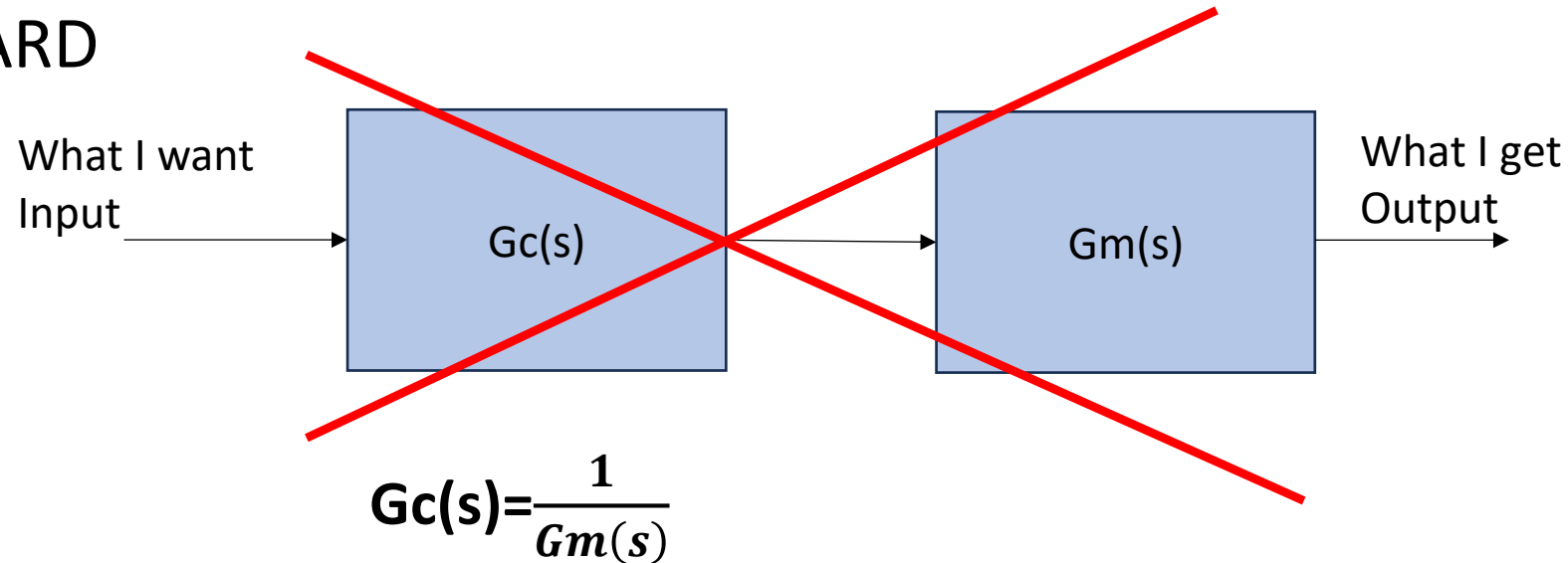
Measure speed (rad/s)



We don't have motor model, we use real motor

What could be an easy way to regulate the speed of my motor?

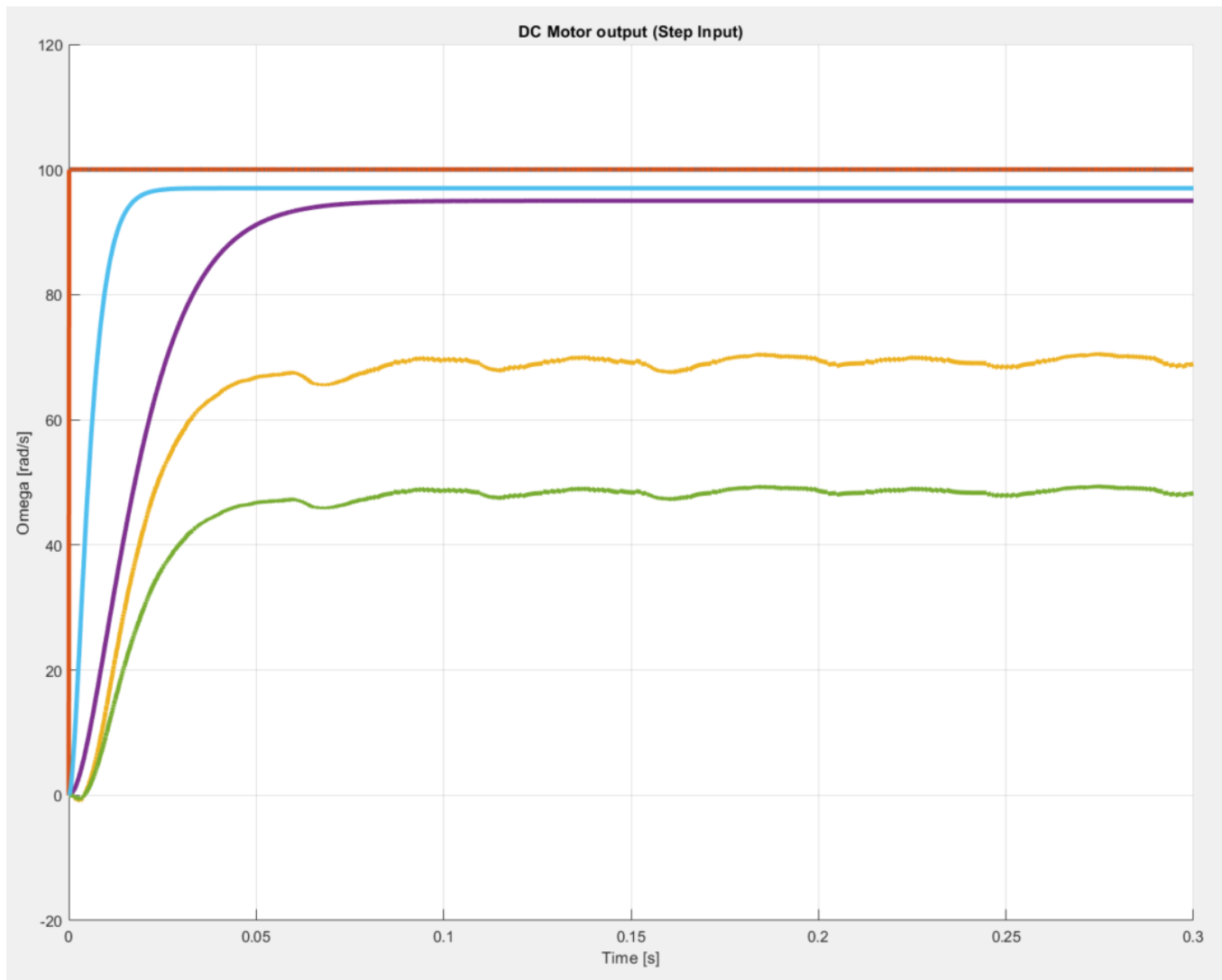
FEEDFORWARD



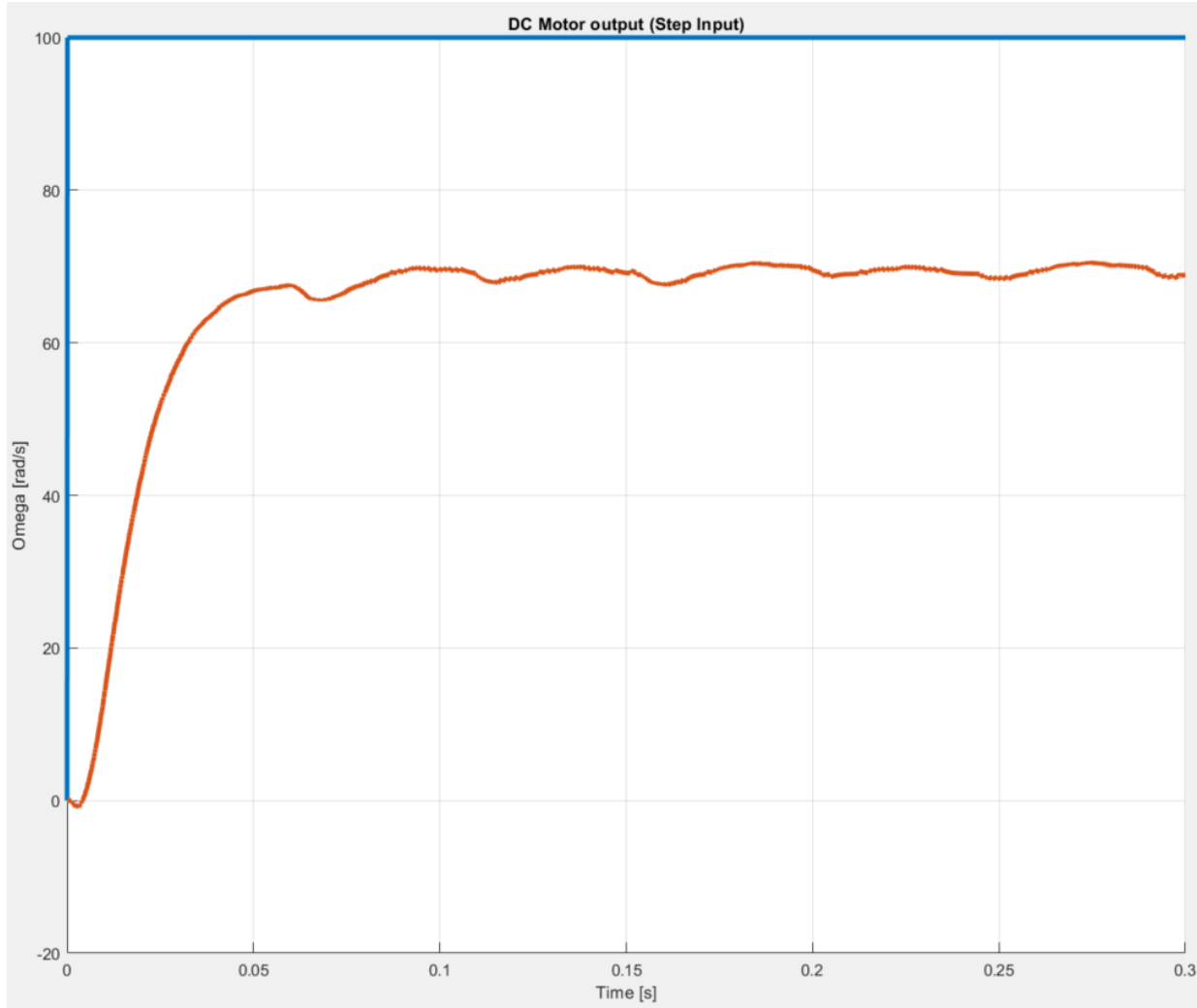
What I want $\longrightarrow$ What I get



Let' have a look in simulink...



What is the real output?



And now, what's happen to your colleague?

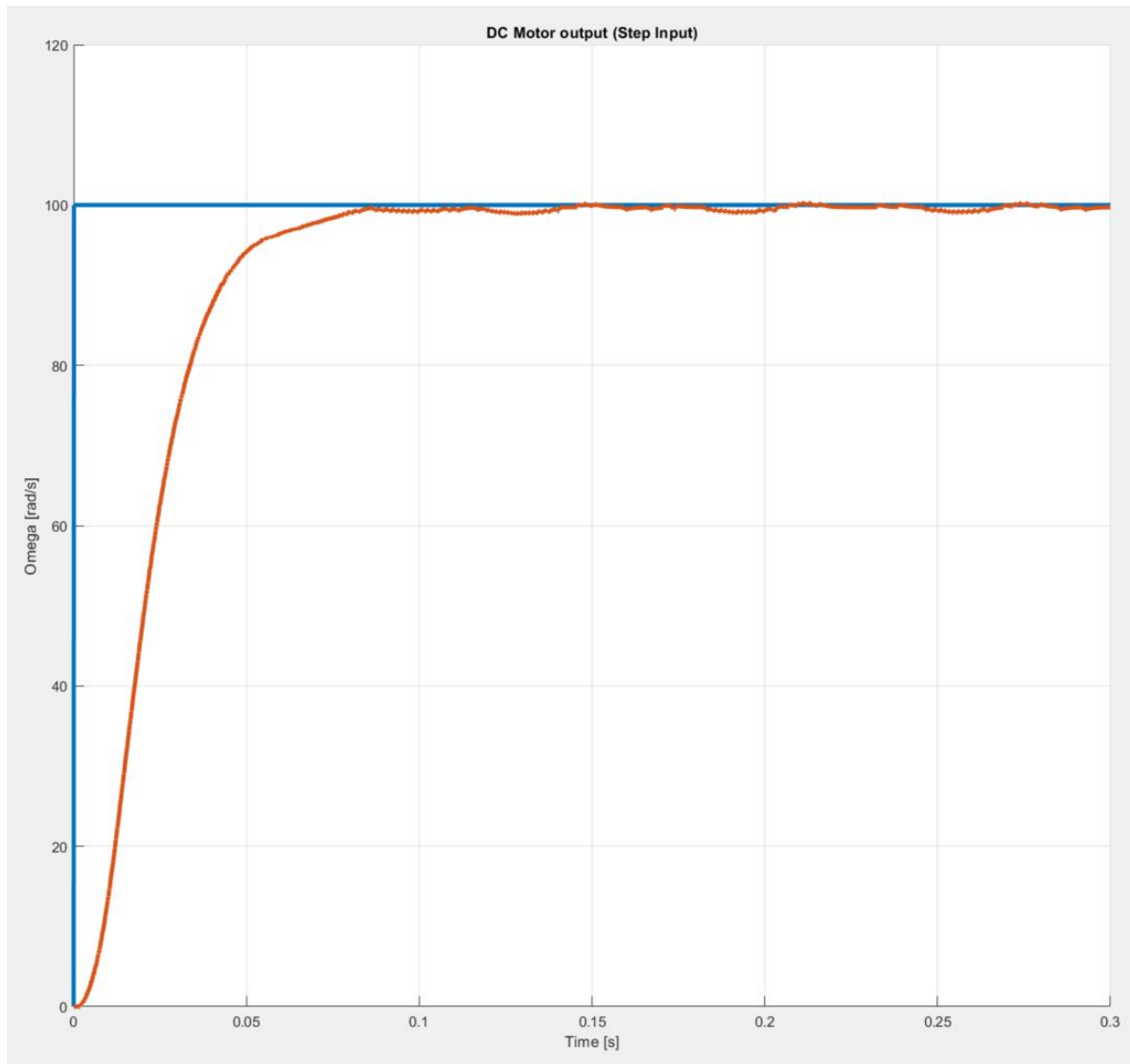
Why do we have this «Big Error»?

- Slower
- steady state error

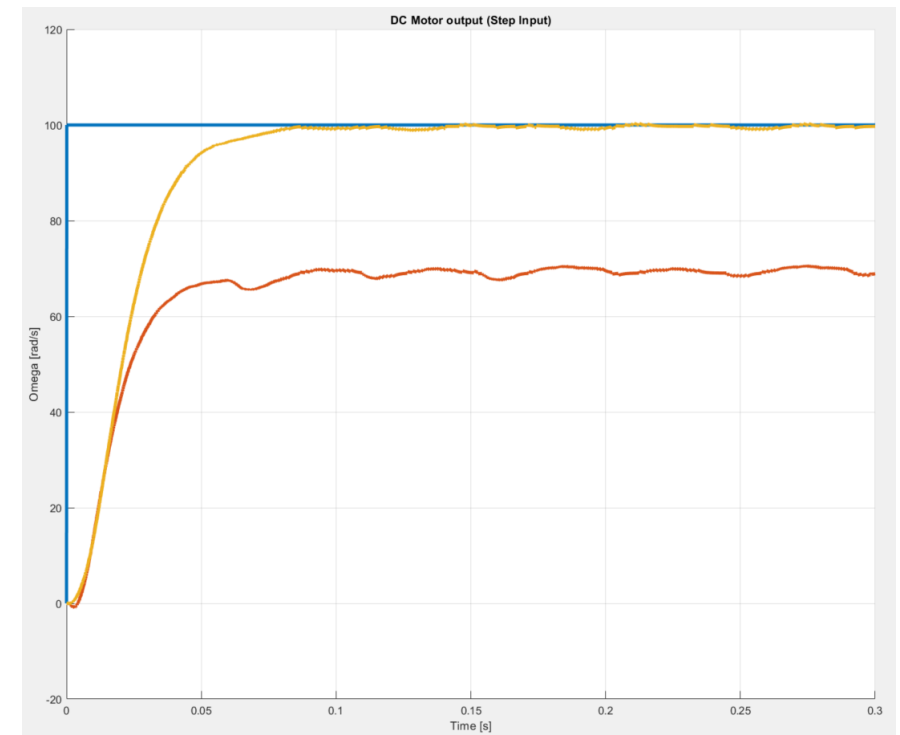


What are the problems?

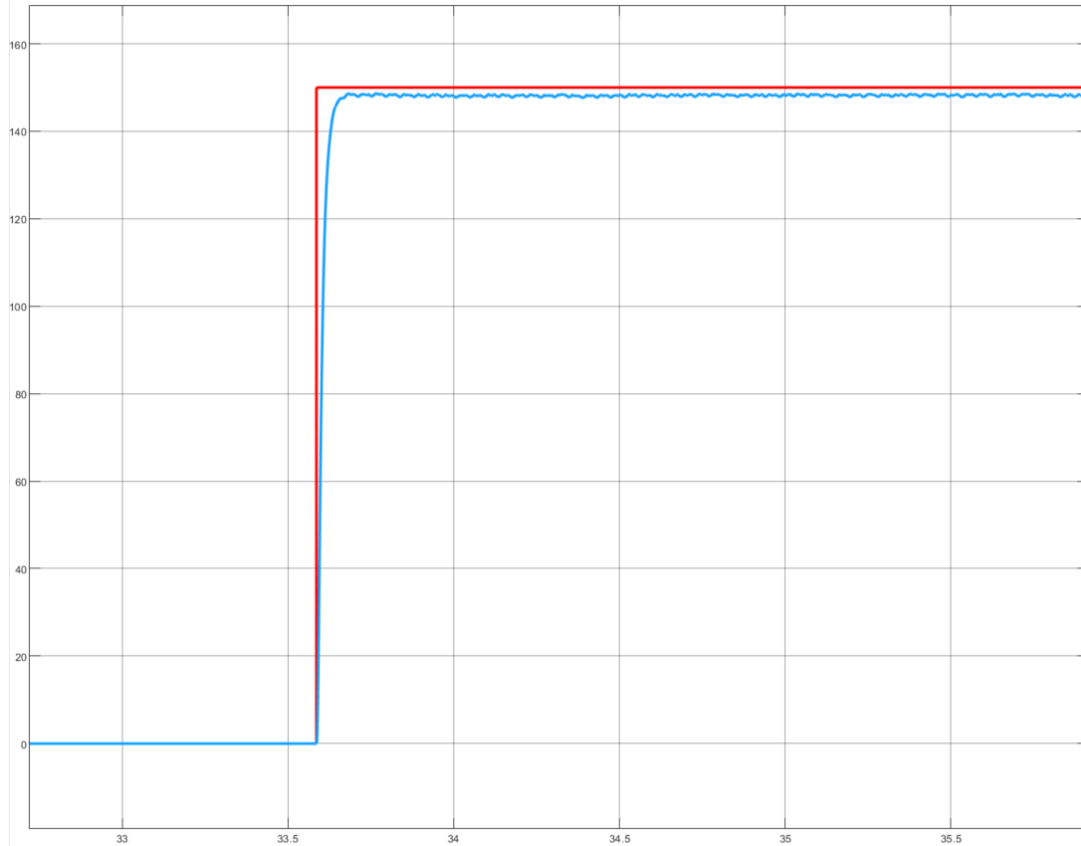
Let's have a look in simulink...



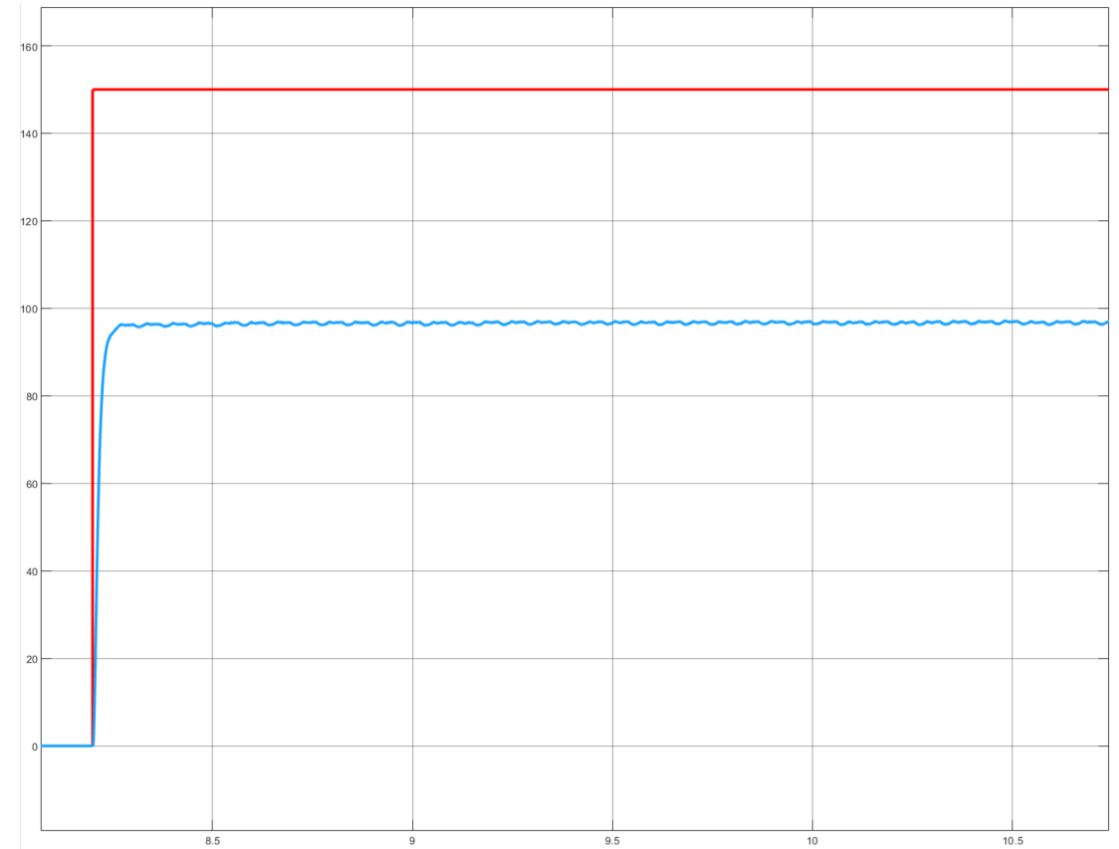
With LS we can do this...



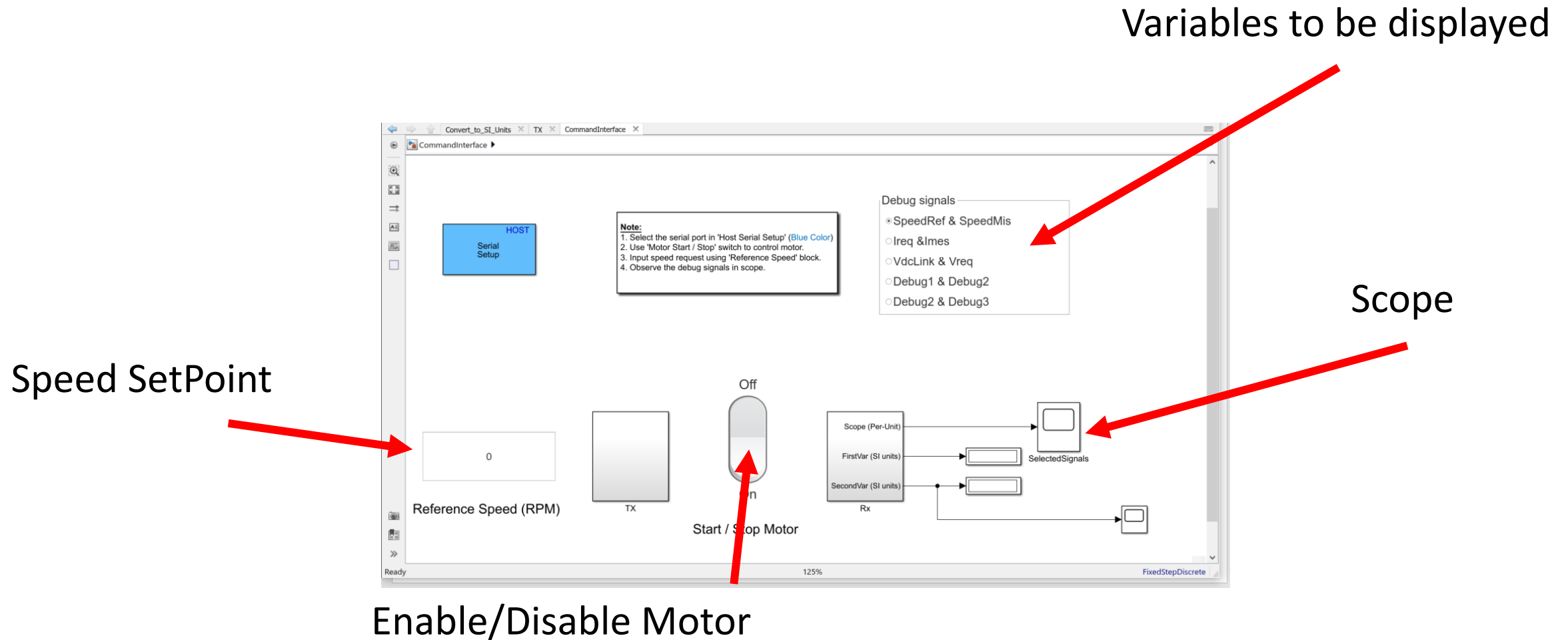
Without LOAD



With LOAD



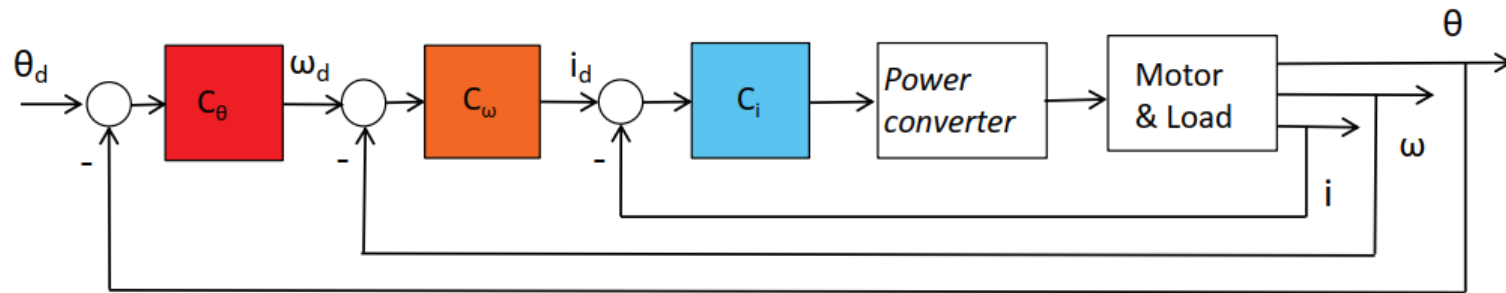
LET'S SEE THE COMMAND INTERFACE



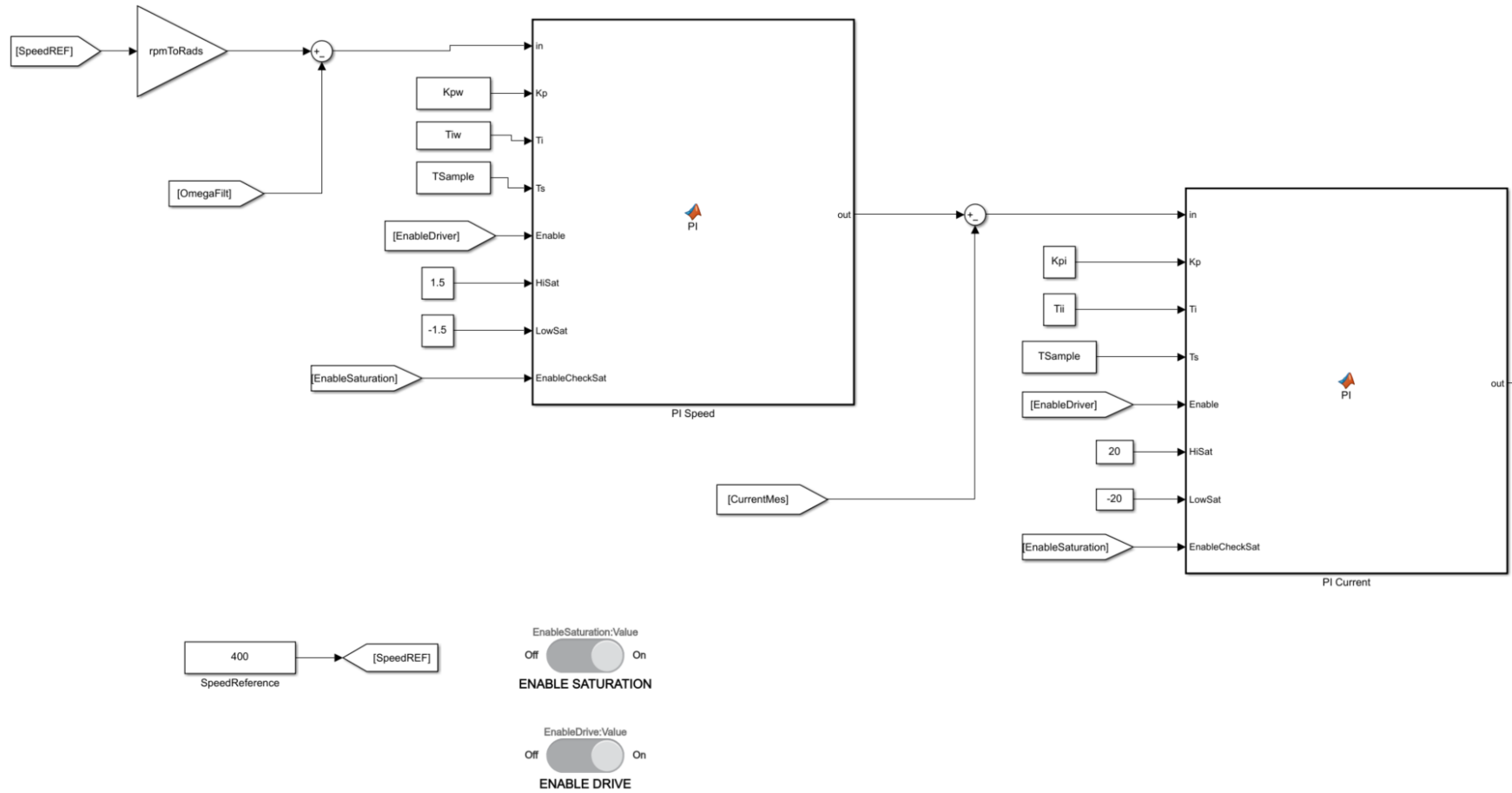
Cascade control of an electric motor

In “high performance” (high dynamics) applications, the control of an electric motor can be “naturally” implemented with a cascade structure, composed of:

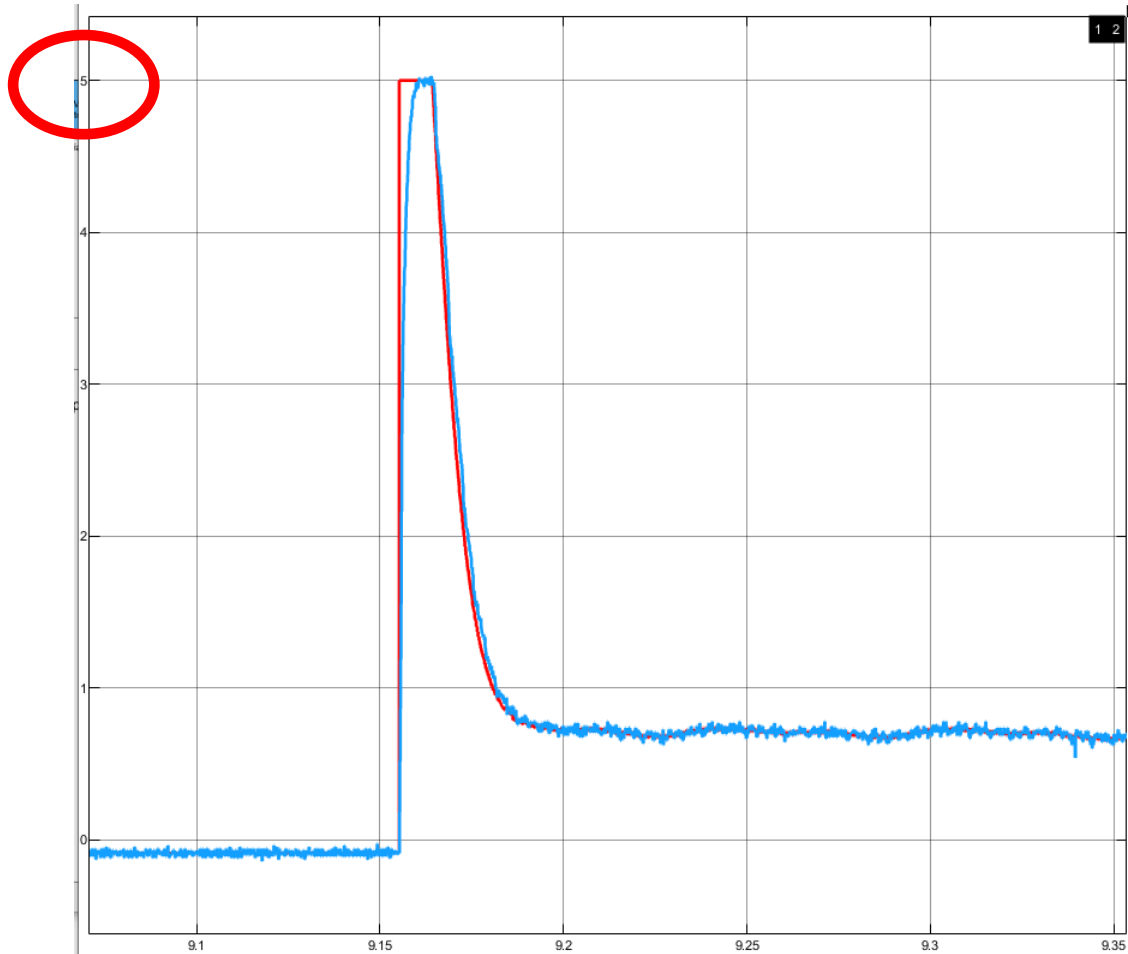
- current control loop, or *torque* control loop
- velocity control loop
- position control loop (not necessarily present)



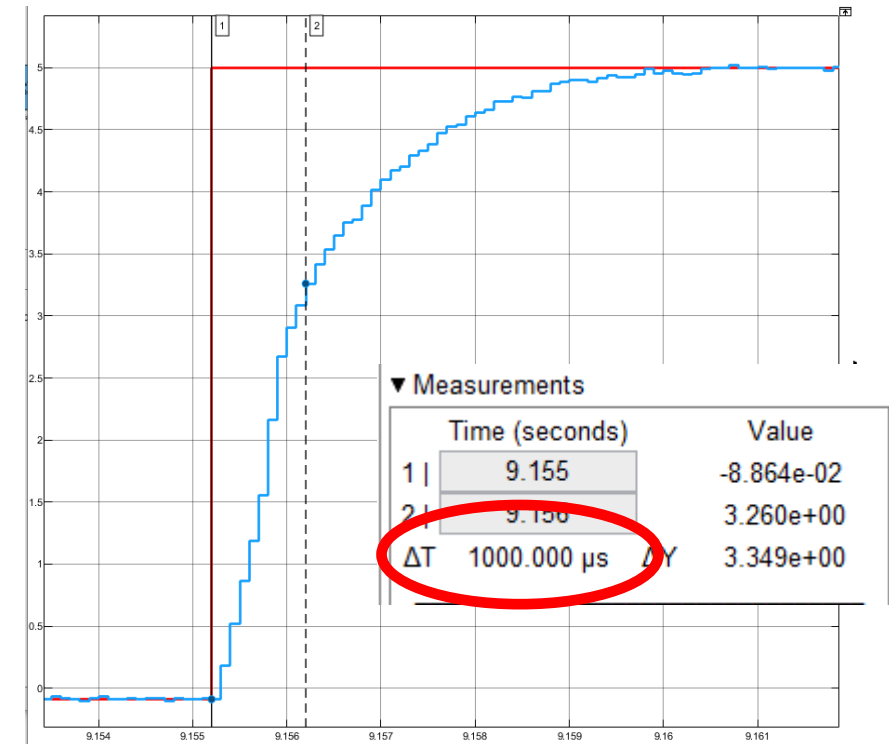
Often, proper *feedforward control actions* are added to the standard feedback loops.



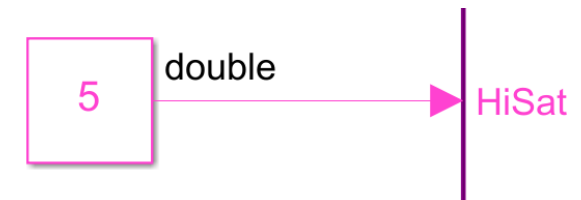
ALCUNI ESEMPI DI TRACING:



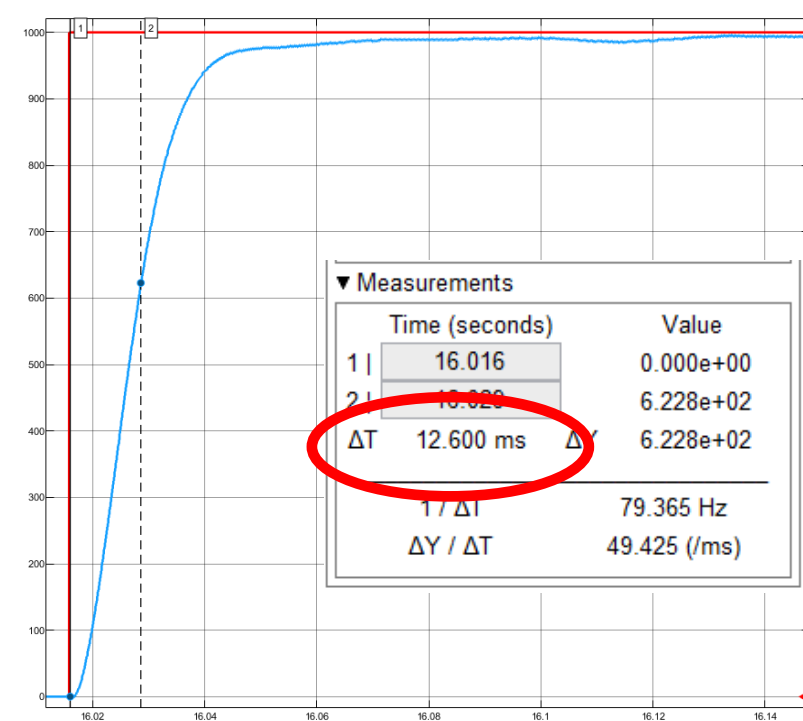
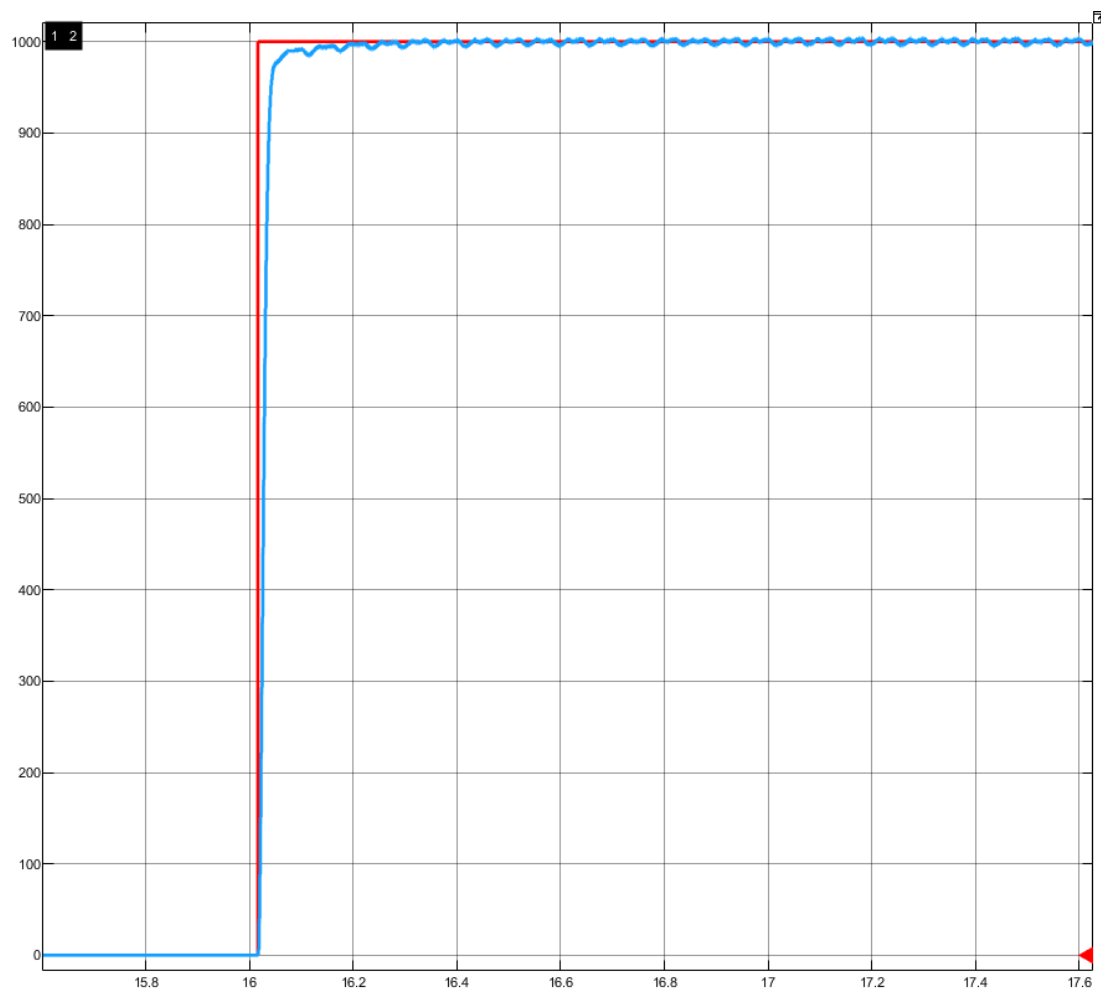
I_{REF}
 I_{MIS}



%Desired Taol
 $Taol=1e-3;$

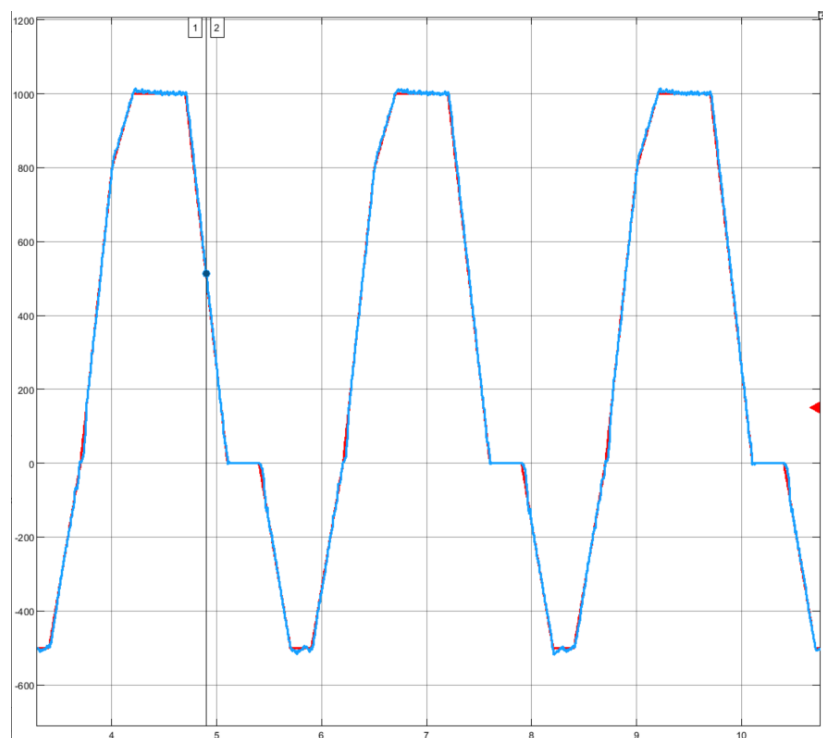


ALCUNI ESEMPI DI TRACING: OmegaREF OmegaMIS

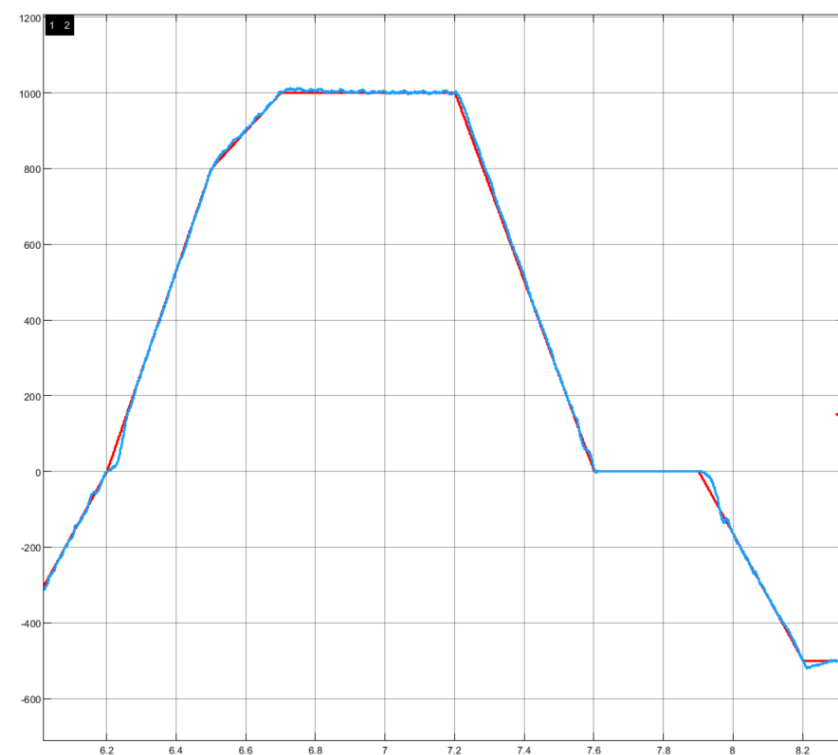


%Desired Taow
Taow=10e-3;

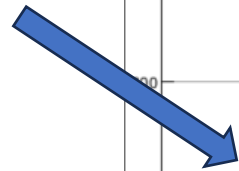
OmegaSetPoint-OmegaMis



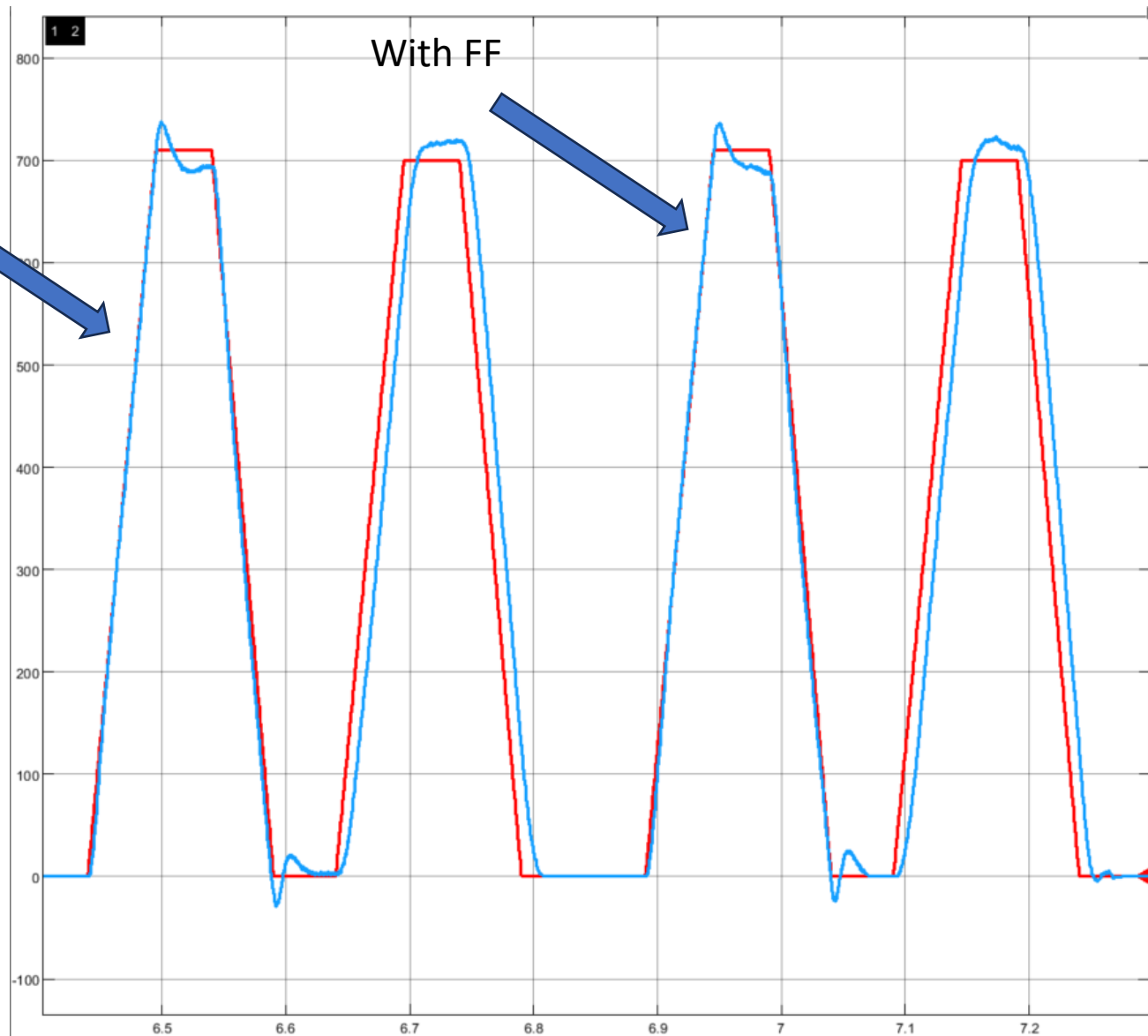
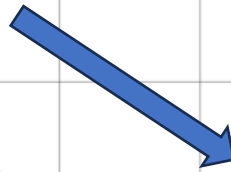
OmegaSetPoint-OmegaMis



With FF



With FF

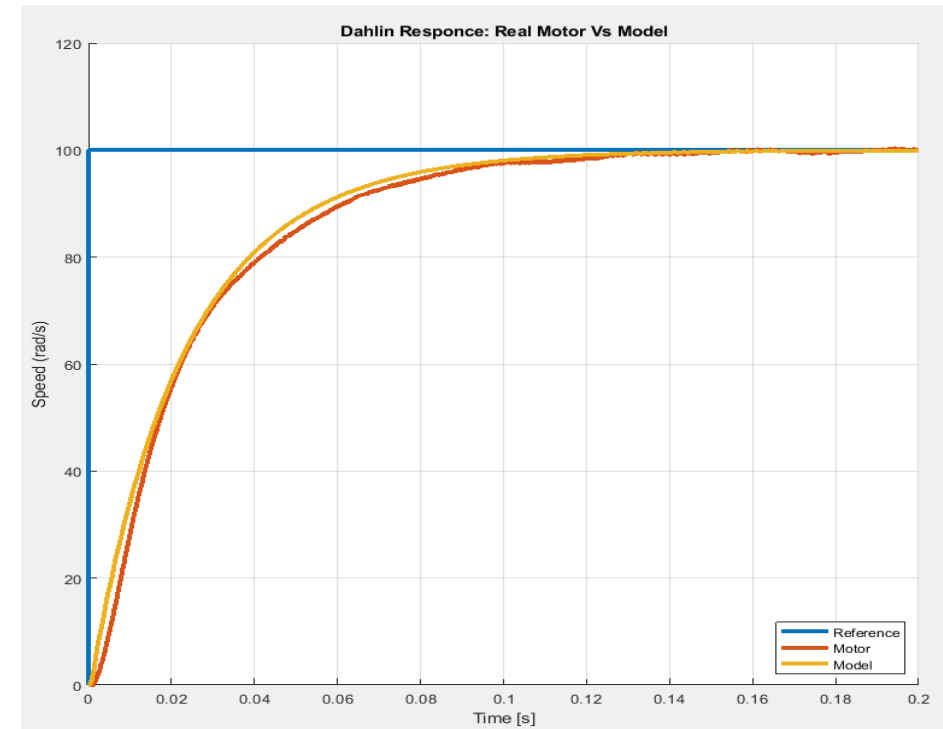
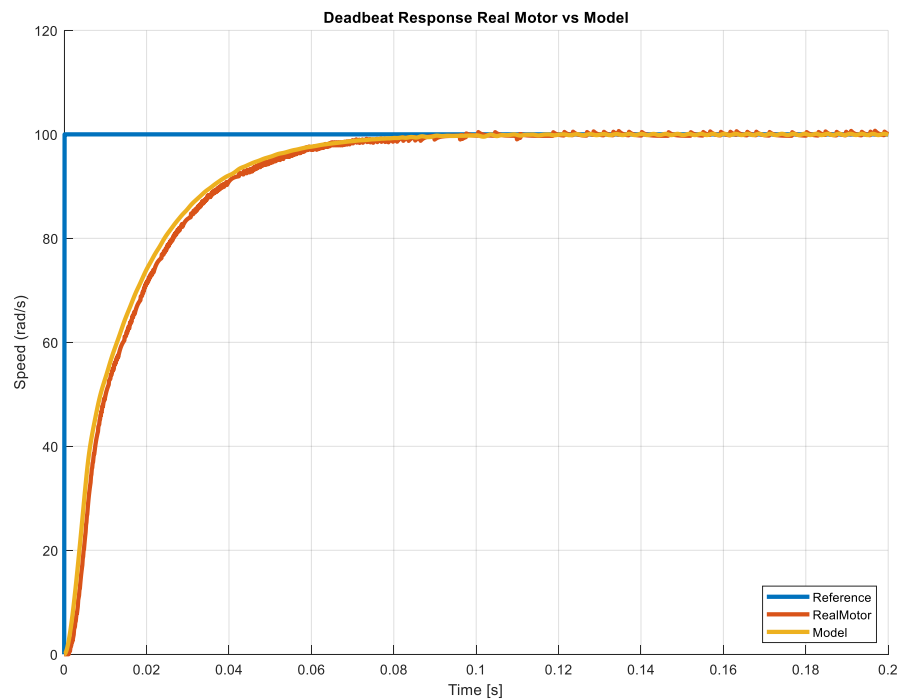


Designing control in discrete-time

Design in discrete-time:

DEADBEAT

DHALIN





Let's have a look in simulink...

Thank you for your attention!